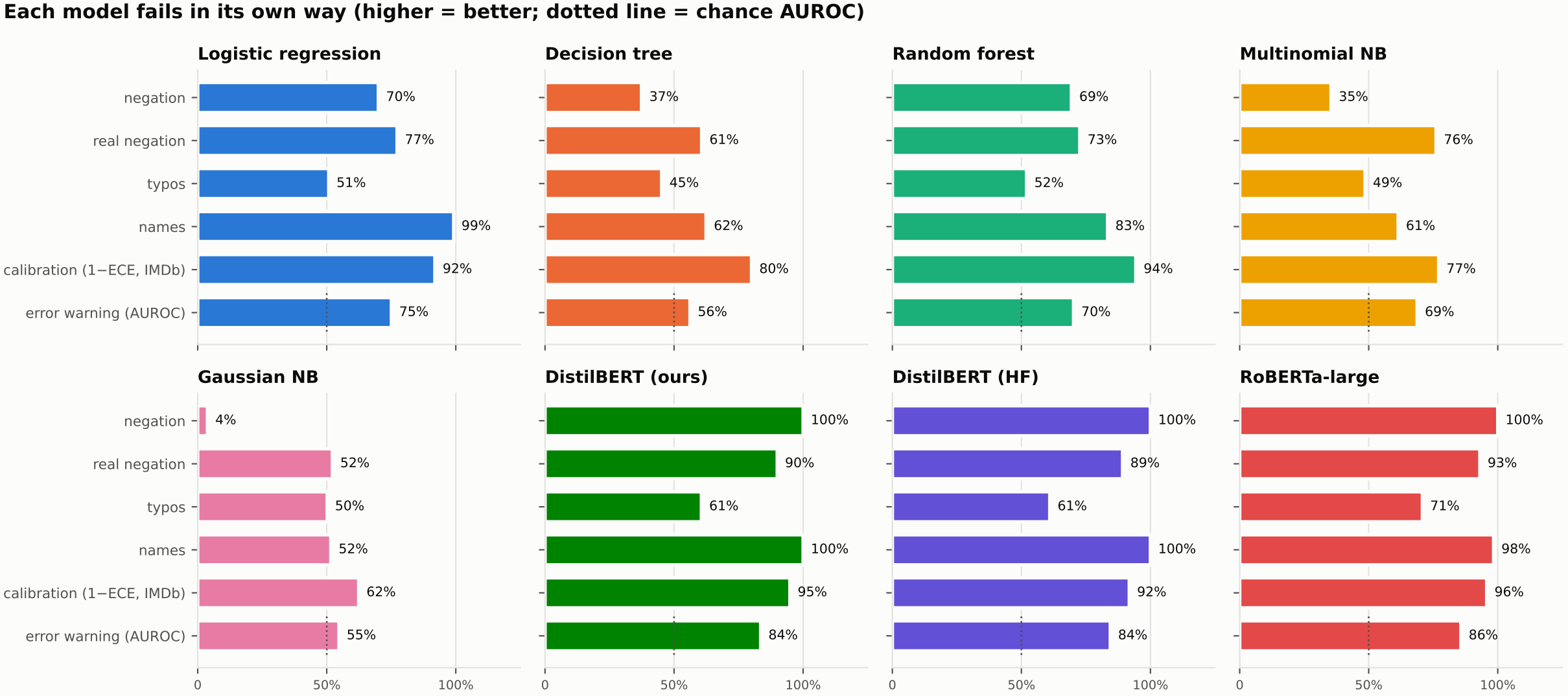
Each model fails in its own way (higher = better; dotted line = chance AUROC)
Logistic regression
Decision tree
Random forest
Multinomial NB
Gaussian NB
DistilBERT (ours)
DistilBERT (HF)
RoBERTa-large
negation
real negation
typos
names
calibration (1−ECE, IMDb)
error warning (AUROC)
70%
77%
51%
99%
92%
75%
37%
61%
45%
62%
80%
56%
69%
73%
52%
83%
94%
70%
35%
76%
49%
61%
77%
69%
4%
52%
50%
52%
62%
55%
100%
90%
61%
100%
95%
84%
100%
89%
61%
100%
92%
84%
100%
93%
71%
98%
96%
86%
0
50%
100%